

ogy engineering discipline of designing and building nanorobots, with devices ranging in size from 0.1µm to 10µm and constructed of nanoscale or molecular components. The names nanobots, nanoids, nanites, nanomachines or nanomites have also been used to describe these devices currently under research and development.

Nanomachines are largely in the R&D phase, but some primitive molecular machines and nanomotors have been tested. The first useful applications of nanomachines might be in medical technology which could be used to identify and destroy cancer cells. Another potential application is the detection of toxic chemicals and the measurement of their concentrations in the environment.

Medical robotics

Medical robotics is causing a paradigm shift in therapy. The rapid growth in medical robotics is driven by a combination of technological improvements (motors, materials and control theory), advances in medical imaging (higher resolutions, magnetic resonance imaging and 3D ultrasound) and an increase in surgeon/patient acceptance of both laparoscopic procedures and robotic assistance. Intensive research on medical robotics in current scenario and future may encroach various medical domains, such as:

Neurological, Orthopedics, General laparoscopy, Percutaneous, Steerable catheters, Radiosurgery, Prosthetics and exoskeletons and Assistive and rehabilitation systems.

Medical robots must develop a firm basis in improved medical outcomes, or risk being displaced by pharmaceuticals, tissue engineering, gene therapy and rapid innovation in manual tools. Robots will see more use for medical-training purposes, bolstered by improved tissue-modelling capabilities, by the increasing objectivity in healthcare assessment, by advances in computer simulations, and as a result of increased data mining arising naturally from improved data connectivity between devices and between institutions. Im-

provements in medical robotics must address and solve real problems in healthcare, ultimately providing a clear improvement in quality of life when compared with the alternatives.

Robots for surveillance and intervention

Surveillance and intervention robots protect homes, public buildings, industrial sites or a country's borders. They generally work on the ground, but may also operate on or under water or in the air. These robots require some cognitive capabilities, particularly with respect to decision making, planning and situation awareness. Currently, their primary task is to gain information and to report back. In the mid term, the use of flying robotic platforms for surveillance and monitoring will increase, in parallel with a maturation of all relevant regulations.

In the long term, such robots will also be able to accomplish more complex tasks such as responding to sudden and unexpected events, and identifying abnormal activities or potentially dangerous situations. Complex security missions will also increasingly require the deployment and cooperation of multiple robotic systems.

Edutainment robots

These robots will interact with humans on a cognitive and physical level. Their task may be to help educate a child, play games with them or provide a social companion for an elderly person. Multi-modal communication including the assessment of a person's emotional state and the physical expression of emotions and gestures are of special importance in this context. Pupils, students and enthusiasts may learn much about technologies related to robotics in the process of building such systems.

India's strength and future planning

Robotics relies on a variety of fundamental domains and is thus to a large extent the science of integrating a broad spectrum of technologies. All technologies essential to robotics have aspects that are almost exclusively relevant in

the context of robotics and aspects that are relevant not only to robotics but also to other domains. Good examples of the first, robotics-driven group are manipulation, navigation and perception. Batteries provide a good example of the second group where advances will benefit robotics, but where, for now, robotics will not be a driving force.

Competitive advantages in high-technology areas are hard won. India must not only retain leadership where this has been achieved, but also take the lead in first-wave technologies. For India's success, it will be vital to capitalise on its existing strong academic base through well-managed technology transfer. However, India cannot afford to only concentrate on areas of strength; it will also need to foster technologies that could become critical barriers to market. In areas of relative weakness, an informed decision has to be made whether a dependence on others is acceptable.

Conclusion

Robotics is the art and commerce of robots, their design, manufacture, application and practical use. Robots will soon be everywhere, in our home and at work. They will change the way we live. This will raise many philosophical, social and political questions that will have to be answered.

With both the government and non-government organisations showing an avid interest in the development of robotics, the future of robotics in India is bright and one day it will be an integrated part of the Indian public serving humanity. But robotics research and education is still at infancy and more areas need to be explored in the near future with intensive training and research at under-graduate, post-graduate and doctoral levels. ●

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